MIL-T-43694(GL)

16 March 1970
SUPERSEDING
(see 6.3)

MILITARY SPECIFICATION

TIEDOWN ASSEMBELIES, FLOOR ANCHORING

1. SCOPE

1.1 This specification covers two types of helicopter floor-achochoring tie-down assemblies. Provision is made for procurement of complete assemblies or certain of the components, as specified (see 6.2).

1.2 Classification. The tiedown assemblies are classified as follows (see 6.2):

Type I - Tiedown Assembly, Part Number 11-1-866
Type II - Tiedown Assembly, Part Number 11-1-719

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	—	Twine, Cotton, Wrapping
UU-P-268	—	Paper, Kraft, Untreated, Wrapping
PPP-B-601	—	Boxes, Wood, Cleated-Plywood
PPP-B-636	—	Box, Fiberboard
PPP-T-45	—	Tape, Gummed, Paper, Reinforce and Plain, For Sealing and Securing

MILITARY

MIL-L-10547	—	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

FSC 1670

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129 - Marking For Shipment And Storage
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

DRAWINGS

ARMY NATICK LABORATORIES

11-1-719 — Tie Down Assembly A/C Floor Anchor, CH-34 Helicopter For Personnel And Cargo Lowering Device
11-1-884 — Loops, Floor Tie Down; Personnel/Cargo Lowering Device, Assembly and Details
11-1-885 — Strap, Tie Down; Assembly and Details
11-1-886 — Tie Down Assembly, Floor Anchoring; Personnel and Cargo Lowering Device.UH-18 And UH-1D Aircraft

(Miniature copies of referenced drawings, identified as figures 1 through 4 respectively are attached for information only.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bid or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

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3. REQUIREMENTS

3.1 First article. - When specified (see 6.2), prior to starting production, the supplier shall submit to or make ready for the contracting officer or his duly authorized representative the required number of samples of the finished item for inspection as specified in 4.2. The preproduction be manufactured in the same facilities and by the production tools and methods to be used for the manufacture of the production items. The approval of the preproduction sample authorizes the supplier to commence production but does not relieve the supplier of responsibility for compliance with all the applicable provisions of this specifications

3.2 Materials and construction. - Materials and construction shall conform to the requirements of the applicable drawing, to sub to subsidiary standards, specifications and drawings applicable thereto, and as specified herein.

3.2.1 Stitches and stitchings. - Thread breaks, bobbin runouts, or two or more consecutive skipped stitches shall be overstitched not less than 1 inch over each end of the defect. Runoffs shall be restitched to conform to the required stitch pattern. Ends of stitching (except for 3.2.1.1) shall be backstitched not less than l/2-inch, except where ends are turned under in a hem or held down by other stitchings. Thread ends shall be trimmed throughout.

3.2.1.1 Stitching, automatic machinery. - Stitching patterns produced by automatic machines may be utilized in lieu of those specified, providing seam strength efficiency and function are equal to, or better than, the specified pattern, and upon approval by the procuring activity. Certification of equivalency, supported by engineering data, may be required.

3.2.3 Identification marking. - Each component shall be marked in accordance with the applicable drawing,

3.3 The workmanship shall conform to the quality established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article. - When required, inspection of the preproduction sample or samples (see 6.2) shall be made of a completely fabricated end item or component for all provisions of this document applicable to end product examination.

4.3 Sampling for inspection and acceptance. - Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspeciton. - In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 In-process inspection. - Inspection shall be made of the end item at appropriate stages of fabrication to determine conformance to the requirements of this document. The Government reserves the right to exclude from consideration for acceptance items for which in-process inspection has indicated nonconformance.

4.4 Inspection for acceptance.-

4.4.1 Inspection lot. - Lots shall be expressed in units of tiedown assemblies or individual components, as applicable.

4.3.2 Examination of end item. - The sample unit for these examinations shall be one tiedown assembly or component, as applicable (see 6.2).

4.4.2.1 Visual examination of assembly or component. - The assembly or component shall be examined for defects classified as follows:

Examine	Defect	classification Major	Minor
Webbing	Not specified type or class	X	
	Cut, tear, holes abrasion or weak area	X	
	Ends not seared or wax dipped as specified		X
	Color not as specified		X
Metal components	Fractured, malformed, bent or otherwise damaged	X	
	Burrs or sharp edges	X	
	Finish not as specified or any corroded area	X	

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Examine	Defect	Classification Major	Minor
Thread (nylon)	Not type, class, or size specified	X	
	Color not as specified		X
Stitches and stitching	Stitching pattern not as specified	X	
	Any row of stitching missing	X	
	One stitch per inch fewer than minimum or more than maximum specified		X
	Two or more stitches per inch fewer than minimum or more than maximum specified	X	
	Thread breaks, bobbin runouts, skipped stitches and runoffs	X	
	Ends of stitching not backstitched or secured as specified	X	
	Tension loose resulting in loose top or bottom thread, or tight tension causing puckering of material at stitch line		X
	Needle chews or bunched stitching	X	
Construction and assembly	Any component missing, not assembled or joined as specified	X	
Marking	Omitted, incorrect, illegible; location, size or color not as specified		X
Workmanship	Grease, oil, ink smears, dirt or other foreign matter; thread ends not trimmed throughout		X

4.4.2.2 Dimensional examination. - The sample unit for this examination shall be one assembly or component, as applicable. The examination shall be made to determine compliance with dimensions specified on the drawings. Any nonconformance shall constitute a defect.

4.4.3 Inspection levels and acceptable quality levels for examination. Inspection levels and acceptable quality levels (AQLs), expressed in defects per hundred units, shall be as follows:

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	Inspection level	AQL
For examination in 4.4.2.1	III	Major 0.25 Total 6.5
For examination in 4.4.2.2	S-3	6.5 One Class
For examination in 4.5	S-2	6.5 One Class
For examination in 4.6	S-2	6.5 One Class

4.3 Examination of preparation for delivery requirements.- An examination shall be made to determine that the packaging, packing and marking complies with section 5 requirements. The sample unit for this examination shall be one shipping container fully prepared for delivery, except that it need not be sealed. Examination for closure defects shall be made of shipping containers fully prepared for delivery. Characteristics to be examined and the defects applicable thereto shall be as specified below.

Examine	Defect
Marking (interior package and exterior container)	Omitted; incorrect; illegible; improper size, location, sequence, improper method of application i.e., not on shipping tag or paper label (when applicable).
Materials	Component missing, damaged, defective, or not as specified.
Workmanship	Improper packaging, or closure of interior containers. i.e., improper closure of container flaps, or improper taping of seams, comers, and manufacturer's joint (as applicable); loose strapping or tape banding; inadequate stapling; bulging or distortion of containers.
Contents (interior package and exterior container)	Number per interior package or exterior container not as specified.

4.6 Examination for palletization. - An examination shall be made to determine that the palletization complies with section 5 requirements. Defects shall be scored in accordance with the list below, The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot.

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Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirements.
Palletization	Not as specified, Pallet pattern not as specified, Interlocking of loads not as specified. Load not bonded with required straps as specified,
Weight	Exceeds maximum load limits.
Marking	Omitted: incorrect, illegible; of improper size, location, sequence or method of application.

5. PREPARATION FOR DELIVERY.

5.1 Packaging. - Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Type I. - Each complete type I tiedown assembly shall be packaged as specified in 5.1.1.1.1, 4.1.1.1.5, and 5.1.1.1.3.

5.1.1.1.1 Tiedown strap - folded in half with the snap hook position on the end opposit the "X" ring. The strap webbing shall be tightly rolled from the snap hook end (snap locking pin captured within roll) and the resultant bundle securely tied with twine confotming to type I or II, 8-ply of T-T-871. Six straps shall then be packaged in an interior bos as hereinafter specified.

5.1.1.2 Tiedown loop. - Each loop shall have the connector snaps and “D” rings positioned in a group at the end opposite the quick fit adapter, The loop webbing shall then be tightly folded around the grouped hardware and the resultant bundle securely tied with twine conforming to type I or II, 8-ply of T-T871. One loop shall then be packaged in an interior box as hereinafter specified.

5.1.1.1.3 Interior box.- Six straps and one loop. bundled and tied as specified in 5.1.1.1.1 and 5.1.1.1.2, shall be compactly packaged together in a snug-fitting fiberboard box conforming to style RSC, grade W6c, of PPP-B-636. Each box shall be closed by means of tape in accordance with the appendix of PPP-B-636.

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5.1.1.1.4 Tiedown strap. - when procured separately (see 6.2d) Each tie-down strap shall be bundled and tied as specified in 5.1.1.1.1.

5.1.1.15 Tiedown loop, when procured separatedly (see 6.2). - Each tie-down loop shall be bundled and tied as specified in 5.1.1.1.2.

5.1.1.2 Type II. Each type II tiedown assembly shall be compactly folded and securely tied at each end with twine conforming to type I or II, 8-ply of T-T-871.

5.1.2 Level C. - Each complete type I tiedown assembly or separately procured (see 6.2) tiedown strap and tiedown loop, and each type II tiedown assembly shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A,B, or C, as specified (see 6.2).

5.2.1 Level A.-

5.2.1.1 Type I. - Four complete type I tiedown assemblies, packaged asas specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC, grade V2s of PPP-B-636; or overseas type style A, 2 load of PPP-B-601, as specified (see 6.2). Each shipping container shall be closed and reinforced in a accordance with the appendix of the applicable contrainer specification.

5.2.1.1.1 Tiedown straps, when procured separately (see 6.2). - Fifty tie-down straps, packaged as specified in 5.1, shall be packed in a snu-fitting shipping container conforming to style RSC-L, grade V2s of PPP-B-636; or overseas type, style A or J, type 2 load of PPP-B-601, as specified (see 6.2). The inside of each fiberboard container shall be fitted with a taped box liner conforming to type CF, class weather-resistant, variety DW, grade V15c, of PPP-B-636. Each fiberboard container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Each fiberboard container shall be closed, waterproofed, and ti reinforced in accordance with the appendix of PPP-B-636. Each wood container shall be waterproofed with a sealed case liner conforming. to type I or II, grade C of MIL-L-10547;and closed and reinforced in accordance with the appendix of PPP-B-601.

5.2.1.1.2 Tied-own loops, when procured separately (see 6.2). - Twelve tie-down loops, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.1.

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5.2.1.2 Type II. - Thirty type II tiedown assemblies packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.1.

5.2.2 Level B. -

5.2.2.1 Type I. - Four complete type I tiedown assemblies, packaged as specified in 4.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1.1 Tiedown straps, when procured separately (see 6.2). - Fifty tiedown straps, packaged as specified in 5.1, shall be packed in asunug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-R-636. The inside of each fiberboard container shall be fitted with a taped box liner conforming to class domestic, variety DW, grade 275 of PPP-B-636. Each fiberboard container shall have the

minimum basis weight kraft paper conforming to grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1.2 Tiedown loops, when procured separately (see 6.2). - Twelve tiedown loops, packaged as specified in 5.1, shall be packed as specified in 5.2.2.1.1.

5.2.2.2 Type II. - Thirty type II tiedown assemblies, packaged as specified in 5.1, shall be packed as specified in 5.2.2.1.1.

5.22.3 When specified (see 6.2), the level B shipping containers for the complete type I tiedown assemblies and separately procured tiedown straps and tiedown loops, and type II tiedown assemblies shall be grade V3c, V3s, or V4s fiberboard boxes fabricted in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification

5.2.3 Level C. - Complete type I tiedown assemblies or separately procured (see 6.2) tiedown straps and tiedown loops, and type II tiedown assemblies, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Palletization. - Unless otherwise specified (see 6.2), complete type I aud type II assemblies only, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be

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bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix ofMIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. - In addition to any special marking required by the contract or order, interior packages, shipping containers, and palletized unit loads (when applicable) shall be marked in accordance with MIL-STD-129. Each unboxed interior package shall have attached a manila colored paper shipping tag for the identification information.

6. NOTES

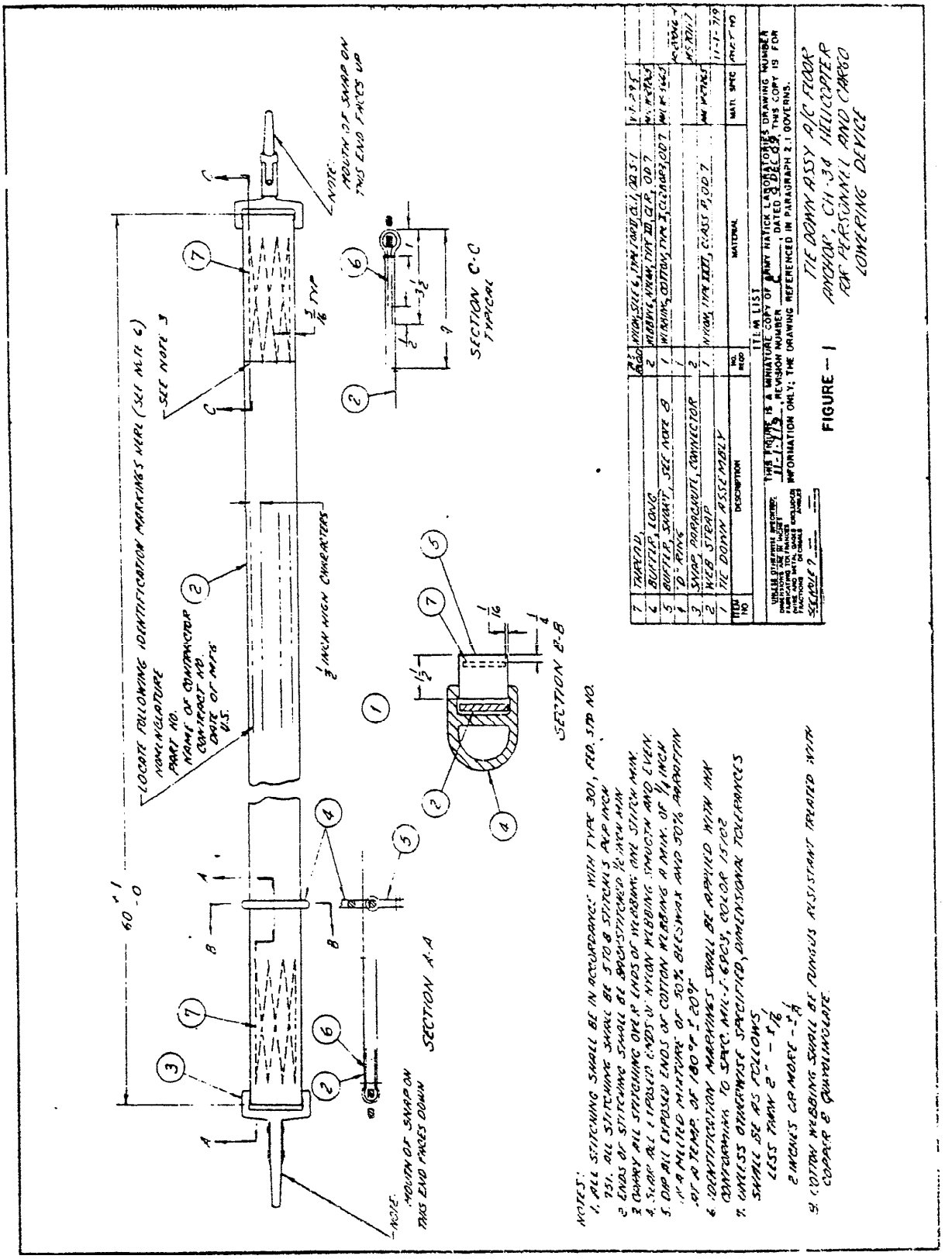
6.1 Intended use. - The tiedowns covered by this specification are utilized as helicopter floor anchoring attachments, in conjunction with a personnel/ cargo lowering device.

6.1.1 Limitations of use. - The use of the tiedowns are restricted to the load capacities and procedures prescribed by TM 10-1670-251-12.

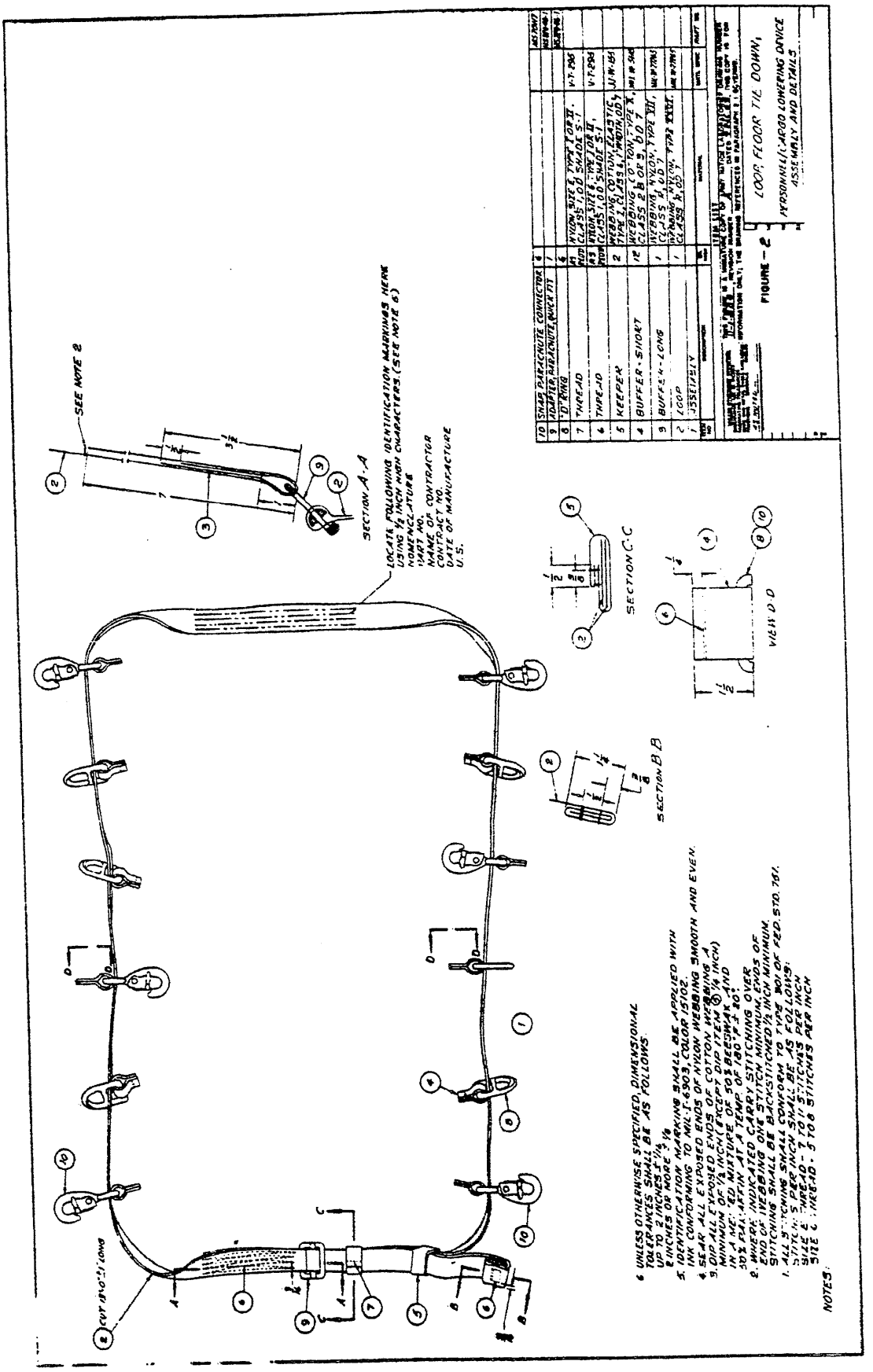
6.2 Ordering data. - Procurement documents should specify the following:

(a) Title and date of this document.
(b) Part number of assembly or component to be procured.
(c) Whether first article is required, and number of samples required (see 3.1).
(d) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2),
(e) When separate procurement of tiedown straps is desired (see 5.1.1.1.4, 5.1.2, 5.2.1.1.1, 5.2.2.1.1 and 5.2.3).
(f) When separate procurement of tiedown loops is desired (see 5.1.1.1.5, 5.1.2, 5.2.1.1.2, 5.2.2.1.2 and 5.2.3).
(g) Type of shipping container desired for level A packing of complete tiedown assemblies or separately procured components (see 5.2.1.1 and 5.2.1.1.1).
(h) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.3).
(i) When palletization is not required (see 5.3).

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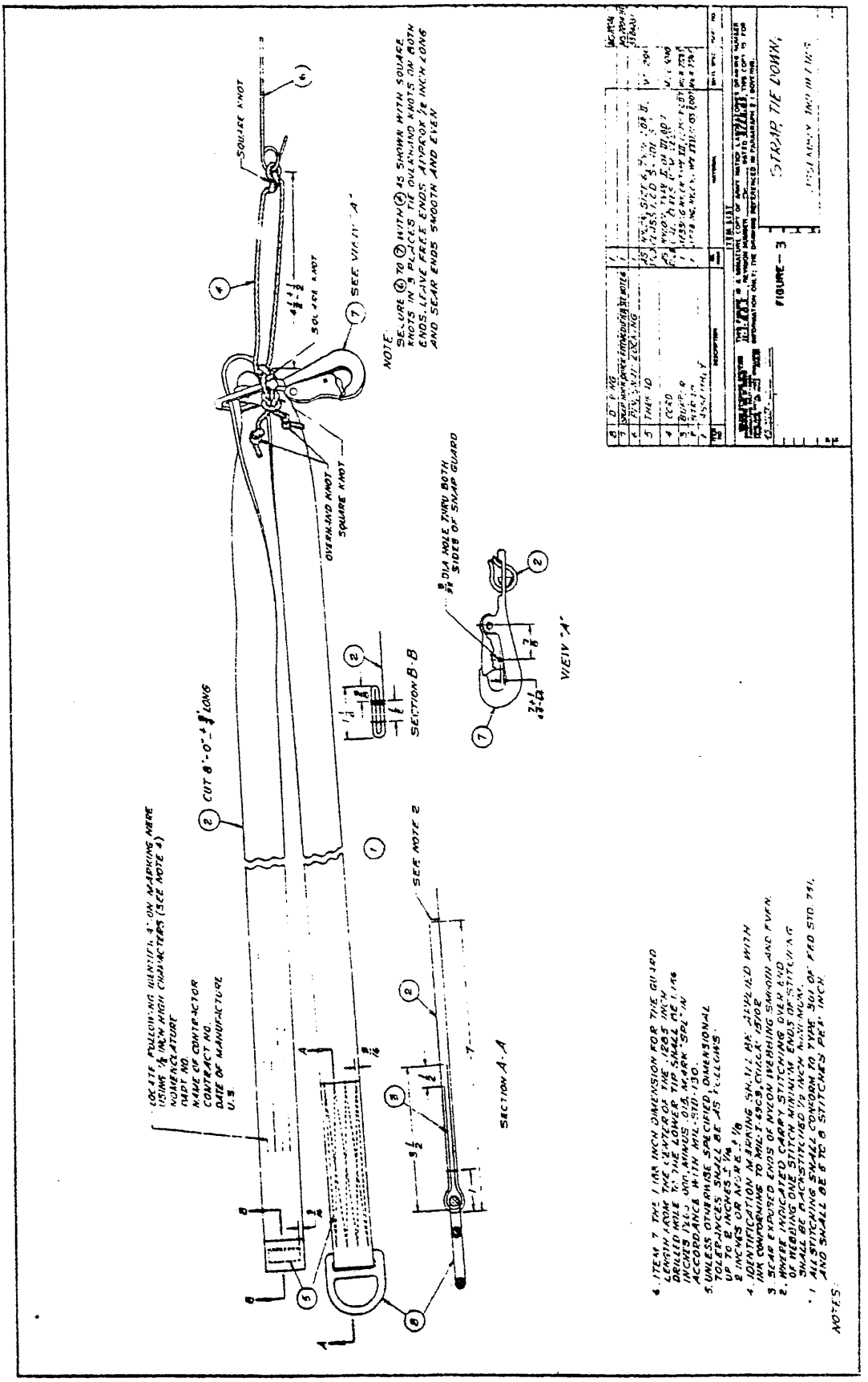
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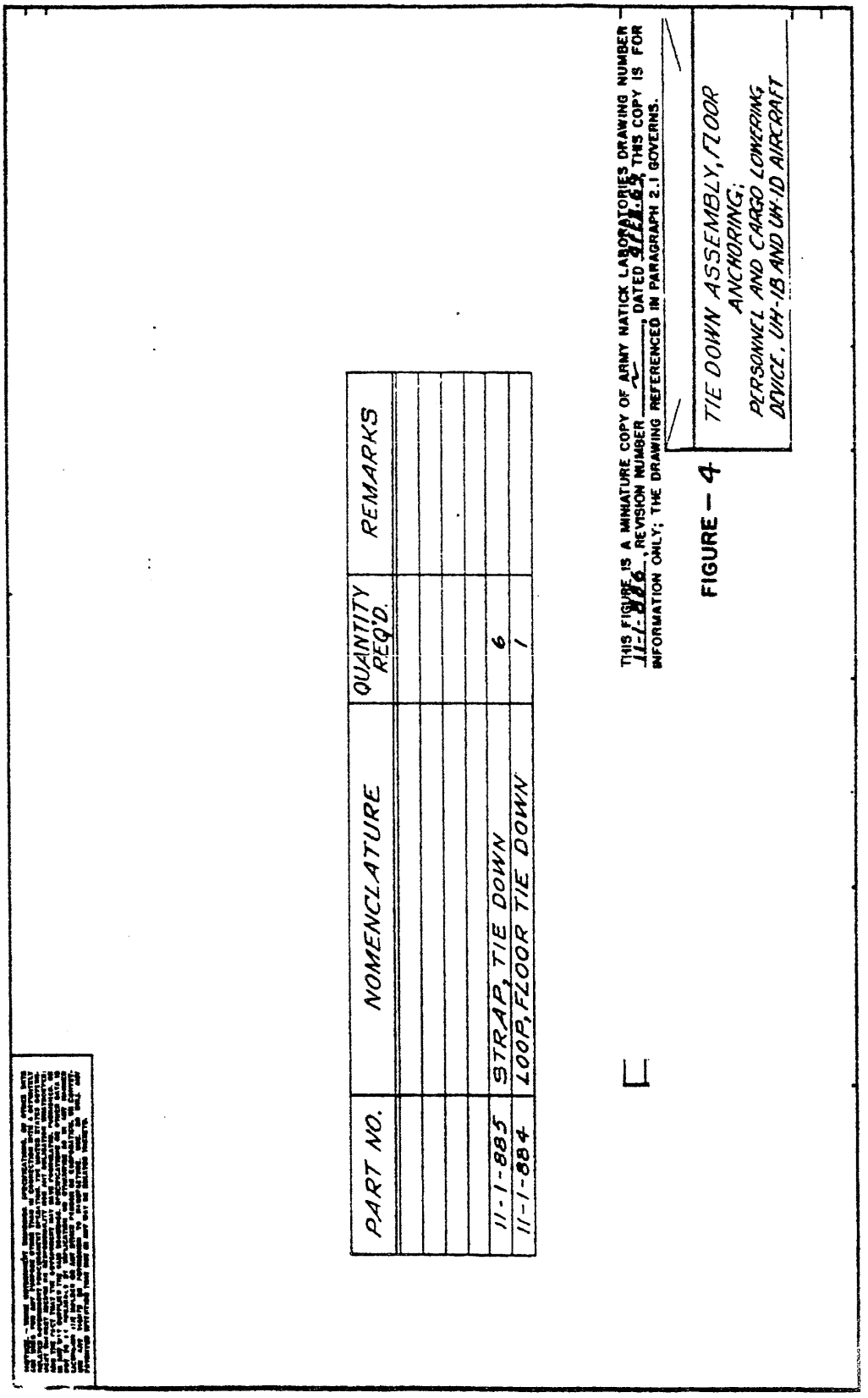
NOTES:
1. ALL STITCHING SHALL CONFORM TO TYPE 301 OF FED. STD. 751.
STITCHES PER INCH SHALL BE AS FOLLOWS:
SIZE 2 THREAD - 5 TO 6 STITCHES PER INCH
SIZE 6 THREAD - 5 TO 8 STITCHES PER INCH
2. WHERE INDICATED CARRY STITCHING OVER END OF WEBBING ONE STITCH AND NOT LESS THAN ½ INCH MINIMUM.
3. IDENTIFICATION MARKING SHALL BE APPLIED WITH INK CONFORMING TO MIL-I-6903, COLOR 15102.
4. SEAR ALL EXPOSED ENDS OF NYLON WEBBING SMOOTH AND EVEN.
5. DIP ALL EXPOSED ENDS OF COTTON WEBBING A MINIMUM OF ¼ INCH (6 MM) EXCEPT DO ITEM 4 IN A MIXTURE OF 50% BEESWAX AND 50% PARAFFIN AT A TEMP OF 180° F ± 20°.
6. UNLESS OTHERWISE SPECIFIED, DIMENSIONAL TOLERANCES SHALL BE AS FOLLOWS:
UP TO 25 INCHES, INCL. ± ½
OVER 25 INCHES ± ¾

10	SNAP PARACHUTE CONNECTOR	4			
9	ADAPTER, PARACHUTE, QUICK FIT	1			
8	D RING	4	NYLON SIZE 1, TYPE I OR II, CLASS 1, OD SHADE S-1	V-T-295	
7	TAPE-AD	AR	NYLON SIZE 1, TYPE I OR II, CLASS 1, OD SHADE S-1	V-T-295	
6	TAPE-AD	AR	NYLON SIZE 1, TYPE I OR II, CLASS 1, OD SHADE S-1	V-T-295	
5	KEEPER	2	WEBBING, COTTON, PLASTIC, TYPE I, CLASS 1, 1¾ INCH WIDTH, OD 7	JJ-W-151	
4	BUFFER-SHORT	12	WEBBING, COTTON, TYPE X, CLASS 2 OR 3, OD 7	MIL-W-5625	
3	BUFFER-LONG	1	WEBBING, NYLON, TYPE XIII, CLASS 1, OD 7	MIL-W-27265	
2	LOOP	1	WEBBING, NYLON, TYPE XIII, CLASS 1, OD 7	MIL-W-27265	
1	ASSEMBLY	1			
NO.	DESIGNATION		MATERIAL		

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PART NO.	NOMENCLATURE	QUANTITY REQ'D.	REMARKS
11-1-885	STRAP, TIE DOWN	6	
11-1-884	LOOP, FLOOR TIE DOWN	1	

THIS FIGURE IS A MINIATURE COPY OF ARMY NATICK LABORATORIES DRAWING NUMBER 11-1-886, REVISION NUMBER ~, DATED 9/11/63. THIS COPY IS FOR INFORMATION ONLY; THE DRAWING REFERENCED IN PARAGRAPH 2.1 GOVERNS.

TIE DOWN ASSEMBLY, FLOOR ANCHORING; PERSONNEL AND CARGO LOWERING DEVICE, UH-1B AND UH-1D AIRCRAFT

FIGURE – 4

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INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426

PREVIOUS EDITION IS OBSOLETE.

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